

# Solutions to Chapter 7 Exercises:

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## Exercise 1: Understanding Federated Learning

Centralized machine learning requires all data to be collected and stored in a central server, where model training is performed. In contrast, federated learning (FL) keeps data on local devices and only shares model updates.

### Advantages of FL in networking:

- Preserves user privacy by avoiding raw data transmission
- Reduces communication overhead
- Enables learning from distributed and heterogeneous data sources
- Supports edge intelligence in IoT and mobile networks

## Exercise 2: Federated Averaging (FedAvg)

Federated Averaging works as follows:

1. A global model is initialized and distributed to devices
2. Each device trains the model locally using its own data
3. Devices send updated model weights to the server
4. The server aggregates updates using weighted averaging

Weighted averaging is used because:

- Devices may have different dataset sizes
- Larger datasets should contribute more to the global model
- It improves convergence and model fairness

## Exercise 3: Impact of Wireless Communication

Wireless conditions significantly affect FL performance:

- **Latency:** Slows down communication rounds and delays convergence
- **Packet Loss:** Leads to missing updates and reduced model quality
- **Limited Bandwidth:** Restricts the number of participating devices

### Example:

In mobile networks, poor channel conditions may prevent some devices from uploading updates, reducing data diversity and slowing learning.

## Exercise 4: Resource Constraints in FL

Main constraints include:

- **Computation:** Limited CPU/GPU affects training speed
- **Memory:** Large models may not fit on edge devices
- **Energy:** Training and communication drain battery

These constraints lead to:

- Slower local training
- Straggler effects
- Reduced participation and convergence speed

#### **Exercise 5: FL in Future Networks**

Federated learning enables AI-native networks by:

- Distributing intelligence across edge devices
- Supporting personalized and adaptive services
- Reducing reliance on centralized data processing

#### **Key challenges:**

- Communication efficiency in wireless environments
- Handling non-IID data across devices
- Security risks (e.g., model poisoning attacks)
- Coordinating large-scale distributed systems